

Ch-1:-

Sets.

Defⁿ:- A Set is a well defined collection of objects.

Note:- (i) Sets are usually denoted by Capital Letters (A, B, C, ...)
(ii) The elements of sets are represented by small letters (a, b, c, ...).

N- Natural, W- Whole, I/Z- Integer, Q- Rational, Q^c- Irrational
R- Real No.

→ If (small) "a" is an element of Set "A", then we say

"a belongs to A"
We write it as " $a \in A$ "

→ If "a" is not an element of Set "A", then we say

"a does not belong to A" → " $a \notin A$ "

Representation of a Set:-

1) Roster or Tabular form:- In it, all the elements of sets being separated by "," & enclosed with "{ }".

eg:- a) Set of all N < 5 ⇒ {1, 2, 3, 4}

b) " " all Vowels ⇒ {a, e, i, o, u}

II) Set Builder form: In it, all the elements of a set possess a single common property which is not possessed by any element outside the set.

eg: a) $\{x | x \text{ is a } N < 5\}$ b) $\{x | x \text{ is vowel in Eng. Alph}\}$

Q. Write the following Set in Roster form:

i) $A = \{x | x \text{ is an integer \& } -3 < x < 7\}$

Sol: $\{-2, -1, 0, 1, 2, 3, 4, 5, 6\}$

ii) $B = \{x | x \text{ is 2-digit } N, \text{ Sum} = 8\}$

Sol: $\{17, 26, 35, 44, 53, 62, 71, 80\}$

iii) $C = \{x | x \text{ is letters in word 'BETTER'}\}$

Sol: $\{B, E, T, R\}$

Note:- The Roster form of a set is unique, but Set builder is not unique.

eg: $\{1, 2\} \rightarrow A = \{x | x \text{ is } N < 3\}$

$B = \{x | x \text{ is the sol}^n \text{ of eqn } (x-1)(x-2)=0\}$

eg: $\{3, 6, 9, 12\} \rightarrow A = \{x | x \text{ is multiple of 3, } < 15\}$

$B = \{x | x = 3n, n \in N < 5\}$

→ Finite Set

Empty Set:- A Set which does not contain any element.

→ Also known as:- Null Set, Void Set.

→ We represent it by $\emptyset$ or $\{\}$ ← empty brackets.

eg:- $\{x | x \in \mathbb{N} \text{ \& } 1 < x < 2\} \rightarrow \{\}$

Finite & Infinite Set:-

Finite:- A Set which is Empty or contain finite no. of element.

Infinite:- " " " don't contain finite no. of element.

eg:- F :- ω be the Set of days of week.

I :- G be the Set of points on the line.

Q. Which are F or I Set:-

i) $\{x | x \in \mathbb{N} \text{ \& } (x-1)(x-2) = 0\} \Rightarrow \text{Finite}$

ii) $\{x | x \in \mathbb{N} \text{ \& } x^2 = 4\} \Rightarrow \text{Finite}$

iii) $\{x | x \in \mathbb{N} \text{ \& } x \text{ is prime}\} \Rightarrow \text{Infinite}$

iv) $\{x | x \in \mathbb{N} \text{ \& } x \text{ is odd}\} \Rightarrow \text{Infinite}$

Equal Set:- Two Sets A & B are said to be equal, if they have exactly same element. And we write $A = B$.

→ Otherwise, the Sets are said to be unequal, write it as $A \neq B$.

eg:- $A = \{1, 2, 3, 4\}$; $C = \{a, b, c, d\}$ } $A = B$; $B = D$; $A = D$
 $B = \{4, 3, 1, 2\}$; $D = \{2, 1, 4, 3\}$ } but $\Rightarrow C \neq A$; $C \neq B$; $C \neq D$.

Quadratic Eqⁿ:- $\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Q. Find pair of equal sets:-

i) $A = \{0\} \Rightarrow \{0\}$
 $B = \{x | x > 15 \& x < 5\} \Rightarrow \{\} \text{ or } \emptyset$
 $C = \{x | x - 5 = 0\} \Rightarrow \{5\}$
 $D = \{x | x^2 = 25\} \Rightarrow \{-5, 5\}$
 $E = \{x | x \text{ is +ve root of eq}^n \Rightarrow \{+5\}$
 $x = +5 \text{ or } -3 \quad x^2 - 2x - 15 = 0 \} \Rightarrow$

ii) A, the set of letters in word 'ALLOY' $\Rightarrow \{A, L, O, Y\}$
 B, " " " " " " 'LOYAL' $\Rightarrow \{L, O, Y, A\}$ equal.

iii) $A = \{n | n \in \mathbb{Z} \& n^2 \leq 4\} \Rightarrow \{-2, -1, 0, 1, 2\}$
 $B = \{x | x \in \mathbb{R} \& x^2 - 3x + 2 \geq 0\} \Rightarrow \{2, 1\}$ not equal.

$0 \in A$ but $0 \notin B \Rightarrow A \neq B$.

Subsets:- A Set 'A' is said to be subset of Set "B", if every element of A is also an element of B.

Symbol
Represented by:- $A \subset B / A \subsetneq B$

eg:- $A = \{1, 2, 3, 4\}$; $B = \{1, 2, 4\} \Rightarrow B \subset A$
 $\rightarrow B \subset A$; if whenever $a \in A$, and $a \in B$

$\rightarrow$ empty set is subset of every set.

$\rightarrow$ In equal set if $A = B \Rightarrow A \subset B$ and $B \subset A$.

$\rightarrow$ Every set is subset of itself.

Q. $\phi, A = \{1, 3\}, B = \{1, 5, 7\}, C = \{1, 3, 5, 7, 9\}$

Insert $\subset$ & $\not\subset$

a) $\phi \subset B$

b) $A \not\subset B$

c) $A \subset C$

d) $B \subset C$

Subsets of Real No.:-

$N = \{1, 2, 3, 4, \dots\}$

$W = \{0, 1, 2, 3, \dots\}$

$Z = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$

$Q = \{x | x = \frac{p}{q} \text{ \& } q \neq 0\}$

$N \subset W$

$W \subset Z$

$Z \subset Q$

$Q \subset R$

$Q^c \subset R$

$N \subset W \subset Z \subset Q \subset R$

Interval as Subsets of real no.:-

a) $(1, 3) = \{x | 1 < x < 3\}$; c) $[1, 3) = \{x | 1 \leq x < 3\}$

b) $[1, 3] = \{x | 1 \leq x \leq 3\}$; d) $(1, 3] = \{x | 1 < x \leq 3\}$

→ Let a, b belong to Real no. and $a < b$,

→ When, the set of real no. is $\{x | a < x < b\}$, called open interval, Denoted as (a, b) .

→ " " " " " " " " $\{x | a \leq x \leq b\}$, close interval, " " $[a, b]$.

→ Semi-Open & Semi-Close $\Rightarrow (a, b], [a, b)$.

Power-Set:- The Set of all possible subsets of a Set 'A' is called, Power-Set.

Denoted by:- $P(A)$.

eg:- $A = \{1, 2, 3\} \Rightarrow \underbrace{\phi, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{2, 3\}, \{1, 3\}, \{1, 2, 3\}}_{\text{Sub sets.}}$

$\{\phi, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{2, 3\}, \{1, 3\}, \{1, 2, 3\}\}$
Power set

eg. $B = \{1, 4\} \Rightarrow \underbrace{\{\emptyset, \{1\}, \{4\}, \{1, 4\}\}}_{\text{Set}} \leftarrow \text{Power set } P(B)$

Cardinality of a Set :- the no. of element in a set A is called Cardinality of that Set A .

Denoted by:- $n(A)$

eg. $A = \{a, b, c, d, e\} \Rightarrow n(A) = 5$

• Note:- Let A be a set with $n(A) = m$, then $n(P(A))$ is 2^m . (See example of Power set).

Universal Set:- Denoted by (usually) $= U$
A set which contain all elements of given Subsets, also called Basic-Set.

Q. Write all subset of following Sets:-
Power set.

i) $\{a\} \Rightarrow \{\emptyset, \{a\}\}$

ii) $\emptyset \Rightarrow \{\emptyset\}$
 $n(P(\emptyset)) = 1$
 $n(\emptyset) = 0$
 $P(\emptyset) = 1$

iii) $\{1, 2, 3, 4\} \Rightarrow \{\emptyset, \{1\}, \{2\}, \{3\}, \{4\}, \{1, 2\}, \{2, 3\}, \{3, 4\}, \{1, 4\}, \{1, 2, 3\}, \{2, 3, 4\}, \{1, 3, 4\}, \{1, 2, 4\}, \{1, 3, 4\}, \{2, 4\}, \{1, 2, 3, 4\}\}$

iv) $\{a, b\} = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$

Q. Write the following as Interval:-

1) $\{x | x \in \mathbb{R}, -4 \leq x \leq 6\} \Rightarrow [-4, 6]$

2) $\{x | x \in \mathbb{R}, -12 \leq x < -10\} \Rightarrow [-12, -10)$

3) $\{x | x \in \mathbb{R}, 0 \leq x < 7\} \Rightarrow [0, 7)$

4) $\{x | x \in \mathbb{R}, 3 \leq x \leq 4\} \Rightarrow [3, 4]$

Q. Write following in Set-builder form:-

- 1) $(-3, 0)$ $\Rightarrow \{x \mid x \in \mathbb{R}, -3 < x < 0\}$
- 2) $[6, 12]$ $\Rightarrow \{x \mid 6 \leq x \leq 12\}$
- 3) $(6, 12]$ $\Rightarrow \{x \mid 6 < x \leq 12\}$
- 4) $[-23, 5)$ $\Rightarrow \{x \mid -23 \leq x < 5\}$

Q. Given $\Rightarrow A = \{1, 3, 5\}$ $B = \{2, 4, 6\}$ $C = \{0, 2, 4, 6, 8\}$

$U = ?$

- a) $\{0, 1, 2, 3, 4, 5, 6\}$ b) $\emptyset$
- ~~c) $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$~~ d) $\{1, 2, 3, 4, 5, 6, 7, 8\}$